Sine function

1 Consider the function $y = \sin x$.

a Complete the table of values giving answers in exact form:

x	0°	30°	90°	150°	180°	210°	270°	330°	360°
$\sin x$									

b Sketch the graph of $y = \sin x$ for $0 \leq x \leq 360$.

c State the sign of $\sin 324^\circ$.

d Which quadrant would the angle 324° lie?

e Describe the trend of the graph of $y = \sin x$.

f State the period of $y = \sin x$.

g State the y-intercept.

h State the maximum y-value.

i State the minimum y-value.

j State the range of the function.

2 Consider the function $y = \sin x$.

a Complete the table of values giving answers in exact form:

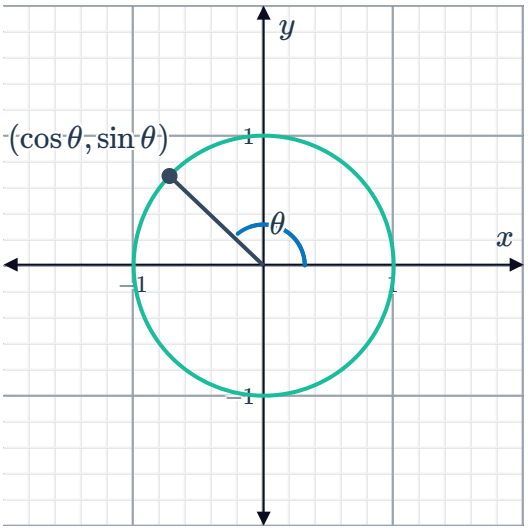
x	0	$\frac{\pi}{6}$	$\frac{\pi}{2}$	$\frac{5\pi}{6}$	π	$\frac{7\pi}{6}$	$\frac{3\pi}{2}$	$\frac{11\pi}{6}$	2π
$\sin x$									

b Sketch the graph of $y = \sin x$ for $0 \leq x \leq 2\pi$.

c State the period of $y = \sin x$.

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3 Given the unit circle, explain why the following statements are true for the graph of $y = \sin\theta$:



a The values of $y = \sin\theta$ lie in the range $-1 \leq y \leq 1$.

b The graph of $y = \sin\theta$ repeats after every 360°.

4 Determine whether the following statements are true of the curve $y = \sin x$:

a The graph of $y = \sin x$ is cyclic.

b As x approaches infinity, the height of the graph approaches infinity.

c The graph of $y = \sin x$ is increasing between $x = -360^\circ$ and $x = -270^\circ$.

d The graph of $y = \sin x$ is symmetric about the line $x = 0$.

e The graph of $y = \sin x$ is symmetric with respect to the origin.

f The y-values of the graph repeat after a period of 180°.

g As x approaches infinity, the graph of $y = \sin x$ stays between $y = -1$ and $y = 1$.

5 Consider the function $y = \sin x$.

a Describe whether the graph is increasing or decreasing in the following intervals:

i $-270^\circ < x < -90^\circ$

ii $-450^\circ < x < -270^\circ$

iii $90^\circ < x < 270^\circ$

iv $-90^\circ < x < 90^\circ$

b Find the x-intercept in the following intervals:

i $0^\circ < x < 360^\circ$

ii $-360^\circ < x < 0^\circ$

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6 Consider the function $y = \sin x$.

a Describe whether the graph is increasing or decreasing in the following intervals:

i $\frac{\pi}{2} < x < \frac{3\pi}{2}$

ii $\frac{3\pi}{2} < x < \frac{5\pi}{2}$

iii $-\frac{3\pi}{2} < x < -\frac{\pi}{2}$

iv $-\frac{5\pi}{2} < x < -\frac{3\pi}{2}$

b Find the x-intercept in the following intervals:

i $0 < x < 2\pi$

ii $-2\pi < x < 0$

7 The number of people visiting Times Square can be modeled by the function

$$P(t) = 24\,000 \sin \frac{\pi}{12} t + 26\,000$$

where $P(t)$ represents the number of people visiting after t hours.

a State the maximum value of this function over one full cycle.

b State the meaning of the answer from part (a) in context.

Cosine function

8 Consider the function $y = \cos x$.

a Complete the table of values giving answers in exact form:

x	0°	60°	90°	120°	180°	240°	270°	300°	360°
$\cos x$									

b Sketch the graph of $y = \cos x$ for $0 \leq x \leq 360$.

c State the sign of $\cos 340^\circ$.

d State the quadrant where an angle with measure 340° lie.

e State the y-intercept.

f State the maximum y-value.

g State the minimum y-value.

h State the domain of the function.

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9 Consider the function $y = \cos x$.

a Complete the table of values giving answers in exact form:

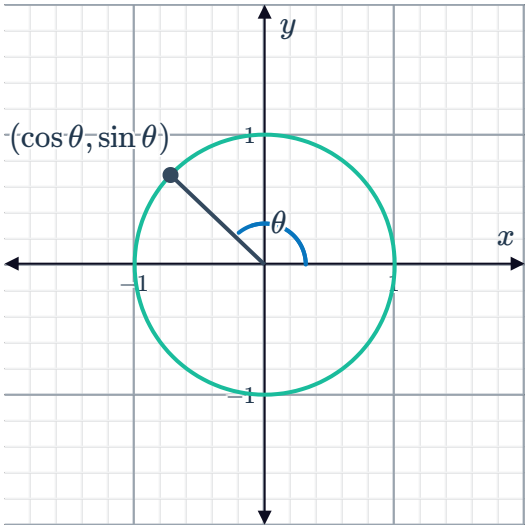
x	0	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	π	$\frac{4\pi}{3}$	$\frac{3\pi}{2}$	$\frac{5\pi}{3}$	2π
$\cos x$									

b Sketch the graph of $y = \cos x$ for $0 \leq x \leq 2\pi$.

10 Given the unit circle, explain why the following statements are true for the graph of $y = \cos\theta$:

a The values of $y = \cos\theta$ lie in the range $-1 \leq y \leq 1$.

b The graph of $y = \cos\theta$ repeats after every 360°.



11 State whether the following statements are true of the graph of $y = \cos x$:

a The graph of $y = \cos x$ is cyclic.

b As x approaches infinity, the height of the graph approaches infinity.

c The graph of $y = \cos x$ is increasing between $x = 90^\circ$ and $x = 180^\circ$.

d The graph of $y = \cos x$ is symmetric about the line $x = 0$.

e The graph of $y = \cos x$ is symmetric with respect to the origin.

f The y-values of the graph repeat after a period of 180°.

g As x approaches infinity, the graph of $y = \cos x$ stays between $y = -1$ and $y = 1$.

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12 Consider the function $y = \cos x$.

a If one cycle of the graph of $y = \cos x$ starts at $x = -90^\circ$, where does the next cycle start?

b Describe whether the graph of $y = \cos x$ is increasing or decreasing in the following intervals:

i $-180^\circ < x < 0^\circ$ **ii** $-360^\circ < x < -180^\circ$

iii $0^\circ < x < 180^\circ$ **iv** $180^\circ < x < 360^\circ$

c Find the x-value of the x-intercepts in the following intervals:

i $0^\circ < x < 360^\circ$ **ii** $-360^\circ < x < 0^\circ$

13 Consider the function $y = \cos x$.

a If one cycle of the graph of $y = \cos x$ starts at $x = -\dfrac{\pi}{2}$, where does the next cycle start?

b Describe whether the graph of $y = \cos x$ is increasing or decreasing in the following intervals:

i $\pi < x < 2\pi$ **ii** $-2\pi < x < -\pi$

iii $0 < x < \pi$ **iv** $-\pi < x < 0$

c Find the x-value of the x-intercepts in the following intervals:

i $0 < x < 2\pi$ **ii** $-2\pi < x < 0$

14 The average monthly temperature of a city can be modeled by a cosine graph. Christa has been living in Brooklyn, New York, where the average annual temperature is 55°F. She would like to move, and live in a location where the average annual temperature is 72°F. When examining the graphs of the average monthly temperatures for various locations, determine whether Christa should focus on each of the following:

a Amplitude **b** Midline **c** Horizontal shift **d** Period

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Tangent function

15 Consider the equation $y = \tan x$.

a Complete the table of values, giving answers in exact form. Write '-' if the value is undefined:

x	0°	60°	90°	120°	135°	180°	225°	270°	300°	360°
$\tan x$										

b Sketch the graph of $y = \tan x$ for $0 \leq x \leq 360$.

c State the y-intercept.

d State the sign of $\tan 345^\circ$.

e State the quadrant where an angle with measure 345° lie.

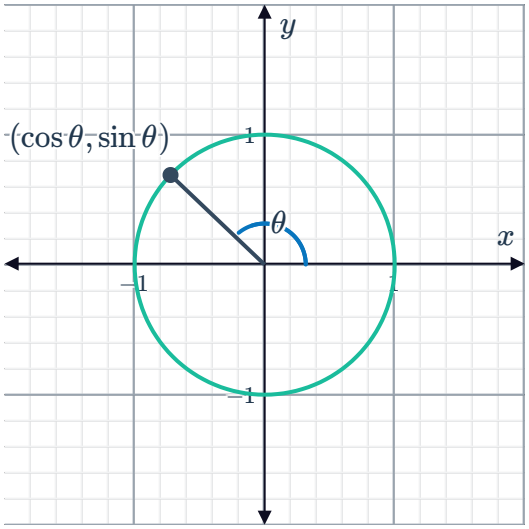
16 Given the unit circle below, state whether the following statements are true of the graph of $y = \tan x$:

a The graph of $y = \tan x$ repeats in regular intervals since the values of $\sin x$ and $\cos x$ repeat in regular intervals.

b The graph of $y = \tan x$ is defined for all values of x.

c Since the radius of the circle is one unit, the value of the function $y = \tan x$ lies in the region $-1 \leq y \leq 1$.

d The range of values of function $y = \tan x$ is $-\infty < y < \infty$.



17 Consider $y = \tan x$ for $-360^\circ \leq x \leq 360^\circ$.

a Determine the period of $y = \tan x$ in degrees.

b Determine the range of $y = \tan x$.

c As x increases, what will be the next asymptote of the graph after $x = 450^\circ$?

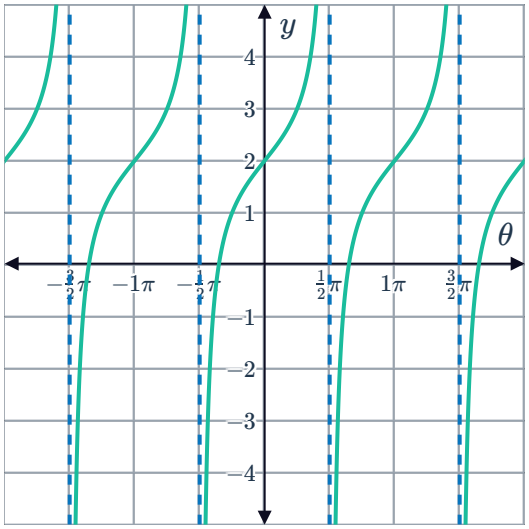
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18 Consider $y = \tan x$ for $-2\pi \leq x \leq 2\pi$.

- **a** State the y-intercept of the graph.
- **b** State the period of the function in radians.
- **c** State the equations of the vertical asymptotes on the domain $0 \leq x \leq 2\pi$.
- **d** Does the graph of $y = \tan x$ increase or decrease between any two successive vertical asymptotes?
- **e** If $x > 0$, find the least value of x for which $\tan x = 0$.

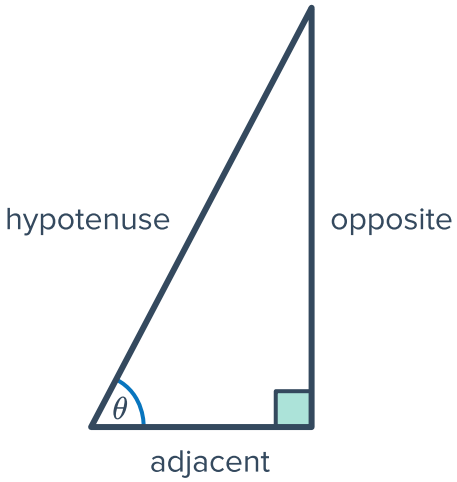
19 Consider the graph of $y = \tan x + 2$.

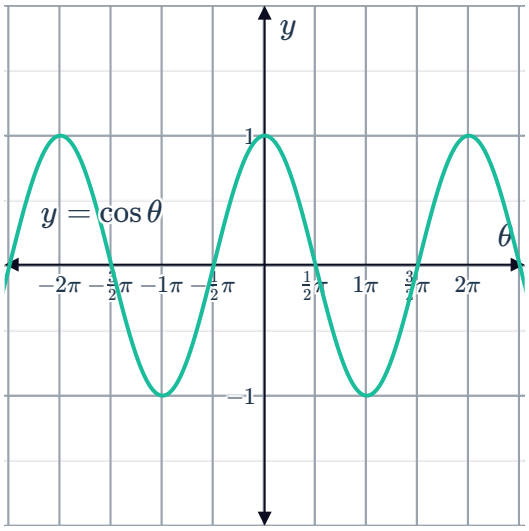
- **a** How long is one cycle of the graph?
- **b** State the x-values for which the equation $\tan x + 2 = 2$, from $x = -2\pi$ to $x = 2\pi$ inclusive.
- **c** State the first positive x-value for which $\tan x + 2 = 3$.
- **d** Using the period of the graph, for what other values of x between $x = -2\pi$ and $x = 2\pi$ does $\tan x + 2 = 3$?
- **e** For what values of x between $x = -2\pi$ and $x = 2\pi$ does $\tan x + 2 = 1$?



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20 Consider the right triangle containing angle θ and the graph of $y = \cos\theta$:





a As angle θ increases from 0 to $\frac{\pi}{2}$, explain what happens to the value of the opposite side in the triangle.

b Now, explain what happens to the value of $\tan\theta$ as angle θ increases from 0 to $\frac{\pi}{2}$, given that $\tan\theta$ is defined as $\frac{\text{opposite}}{\text{adjacent}}$.

c Express $\tan\theta$ in terms of $\sin\theta$ and $\cos\theta$.

d Determine the values of θ for which $\cos\theta = 0$, given $-2\pi \leq \theta \leq 2\pi$.

e Now, state the values of θ between -2π and 2π for which $\tan\theta$ is undefined.

f Complete the tables below:

θ	-2π	$-\frac{7\pi}{4}$	$-\frac{5\pi}{4}$	$-\pi$	$-\frac{3\pi}{4}$	$-\frac{\pi}{4}$
$\tan\theta$						

θ	0	$\frac{\pi}{4}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{7\pi}{4}$	2π
$\tan\theta$							

g Now, sketch the graph of $y = \tan\theta$ on the domain $-2\pi \leq \theta \leq 2\pi$.

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